

# Dynamic analysis of beams, frames and grids

## Special structural elements

---

ME473 Dynamic finite element analysis of structures

Stefano Burzio

2025



## Where do we stand?

| Week | Module                      | Lecture topic                      | Mini-projects       |
|------|-----------------------------|------------------------------------|---------------------|
| 1    | Linear elastodynamics       | Strong and weak forms              |                     |
| 2    |                             | Galerkin and FE methods            |                     |
| 3    |                             | Systematisation of the procedure   | Groups formation    |
| 4    |                             | 3d elements, numerical integration | Project 1 statement |
| 5    | Special structural elements | Bars and trusses                   |                     |
| 6    |                             | Beams, frames and grids            |                     |

## Summary

- Recap week 5
- Euler-Bernoulli planar beams
- Timoshenko planar beams
- Plane frames
- Plane grids
- Three-dimensional frames

## Recommended readings

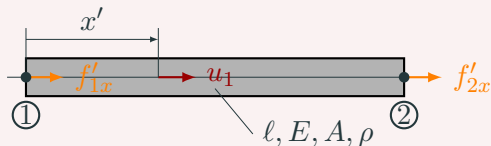
- ① Logan, A first course in the finite element method, 6th ed. (chap. 4 and 5)
- ② Paz and Leigh, Structural dynamics, 6th ed. (chap. 10, 11, 12 and 13)
- ③ Ferreira and Fantuzzi, MATLAB Codes for Finite Element Analysis, 2nd ed. (chap. 6, 7, 8, 9 and 10)

Recap week 6

Vibrations of bars and trusses

---

## Non-oriented bar element



Differential equation governing the dynamics:

$$EA \partial_{x'x'}^2 u_1(x', t) = \rho A \ddot{u}_1(x', t)$$

- Displacements approximation in local coordinate:

$$u_1^h(x', t) = \mathbf{H}(x') \mathbf{q}_{loc}(t) = \begin{bmatrix} h_1(x') & h_2(x') \end{bmatrix} \begin{bmatrix} q'_{1x}(t) \\ q'_{2x}(t) \end{bmatrix}$$

- Linear local shape functions:

$$h_1(x') = 1 - \frac{x'}{\ell} \quad \text{and} \quad h_2(x') = \frac{x'}{\ell}$$

- Semi-discrete weak form:

$$\delta \mathbf{q}_{loc}^T [\mathbf{M}_{loc} \ddot{\mathbf{q}}_{loc}(t) + \mathbf{K}_{loc} \mathbf{q}_{loc}(t) - \mathbf{f}_{loc}(t)] = 0$$

## Non-oriented bar element discretization

- Element stiffness matrix in local coordinates:

$$\mathbf{K}_{loc} = \int_0^\ell EA \frac{d\mathbf{H}^T}{dx'} \frac{d\mathbf{H}}{dx'} dx' = \int_0^\ell EA \begin{bmatrix} (h'_1)^2 & h'_1 h'_2 \\ h'_2 h'_1 & (h'_2)^2 \end{bmatrix} dx' = \frac{EA}{\ell} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

- Element consistent mass matrix in local coordinates:

$$\mathbf{M}_{loc} = \int_0^\ell \rho A \mathbf{H}^T \mathbf{H} dx' = \int_0^\ell \rho A \begin{bmatrix} (h_1)^2 & h_1 h_2 \\ h_2 h_1 & (h_2)^2 \end{bmatrix} dx' = \frac{\rho A \ell}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

- Element applied loads vector in local coordinates:

$$\mathbf{f}_{loc}(t) = \begin{bmatrix} h_1(0) \\ h_2(0) \end{bmatrix} f'_{1x}(t) + \begin{bmatrix} h_1(\ell) \\ h_2(\ell) \end{bmatrix} f'_{2x}(t) = \begin{bmatrix} f'_{1x}(t) \\ f'_{2x}(t) \end{bmatrix}$$

# Arbitrarily oriented bar element

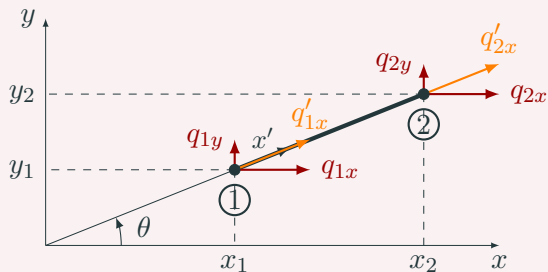
- Displacements in local coordinates:

$$\mathbf{q}_{loc} = [q'_{1x}, q'_{2x}]^T$$

- Displacements in global coordinates:

$$\mathbf{q} = [q_{1x}, q_{1y}, q_{2x}, q_{2y}]^T$$

- Relation between local and global displacements:



$$\underbrace{\begin{bmatrix} q'_{1x} \\ q'_{2x} \end{bmatrix}}_{\mathbf{q}_{loc}} = \underbrace{\begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 & 0 \\ 0 & 0 & \cos(\theta) & \sin(\theta) \end{bmatrix}}_{\mathbf{T}} \underbrace{\begin{bmatrix} q_{1x} \\ q_{1y} \\ q_{2x} \\ q_{2y} \end{bmatrix}}_{\mathbf{q}}$$

## Discretization of arbitrarily oriented bar

- Element stiffness matrix in global coordinates:

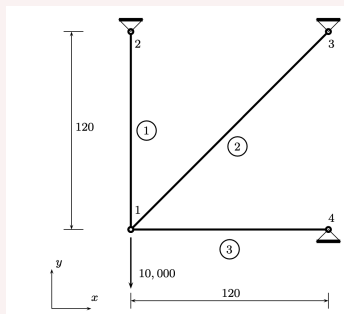
$$\mathbf{K} = \mathbf{T}^T \mathbf{K}_{loc} \mathbf{T} = \frac{EA}{\ell} \begin{bmatrix} \cos^2(\theta) & \sin(\theta) \cos(\theta) & -\cos^2(\theta) & -\sin(\theta) \cos(\theta) \\ & \sin^2(\theta) & -\sin(\theta) \cos(\theta) & -\sin^2(\theta) \\ & & \cos^2(\theta) & \sin(\theta) \cos(\theta) \\ \text{Symm.} & & & \sin^2(\theta) \end{bmatrix}$$

- Element consistent mass matrix in global coordinates:

$$\mathbf{M} = \mathbf{T}^T \mathbf{M}_{loc} \mathbf{T} = \frac{\rho A \ell}{6} \begin{bmatrix} 2 \cos^2(\theta) & 2 \sin(\theta) \cos(\theta) & \cos^2(\theta) & \sin(\theta) \cos(\theta) \\ & 2 \sin^2(\theta) & \sin(\theta) \cos(\theta) & \sin^2(\theta) \\ & & 2 \cos^2(\theta) & 2 \sin(\theta) \cos(\theta) \\ \text{Symm.} & & & 2 \sin^2(\theta) \end{bmatrix}$$

## Illustrative example

**Plane truss:** structure composed of oriented bar elements that all lies in a common plane and are connected by frictionless pins.



| Elements | Nodes | $^e\theta$ | $^e\ell$         |
|----------|-------|------------|------------------|
| 1        | 1, 2  | $90^\circ$ | 120 mm           |
| 2        | 1, 3  | $45^\circ$ | $120\sqrt{2}$ mm |
| 3        | 1, 4  | $0^\circ$  | 120 mm           |

Elementary stiffness and mass matrices:

$$\begin{aligned} {}^e\mathbf{K} &= {}^e\mathbf{T}\mathbf{K}_{loc}{}^e\mathbf{T} \\ {}^e\mathbf{M} &= {}^e\mathbf{T}\mathbf{M}_{loc}{}^e\mathbf{T} \end{aligned} \quad e = 1, 2, 3$$

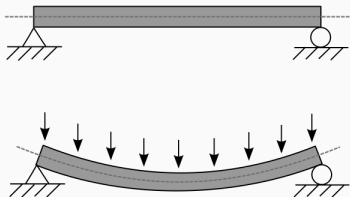
Global stiffness and mass matrices:  $\mathbf{K} = \mathbf{A}_{e=1}^3 {}^e\mathbf{K}$  and  $\mathbf{M} = \mathbf{A}_{e=1}^3 {}^e\mathbf{M}$ .

# Euler-Bernoulli planar beams

---

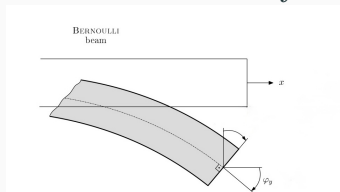
# What is a beam?

- Considered to be a uniaxial (slender) element: the longitudinal direction is sufficiently larger than the other two.
- Cross-section does not change along the element's length.
- Subjected to **transversal loading** that produced significant bending effects.
- Very common type of structures used in steel buildings, bridges, towers, etc...



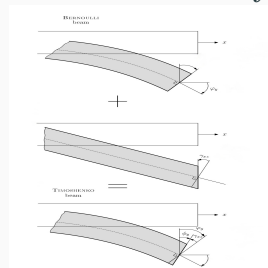
# Euler-Bernoulli and Timoshenko beam assumptions

## Thin beam theory



- Cross-sections remain **planar** and **perpendicular** to the longitudinal axis after bending.
- Shear deformation  $\varepsilon_{12}$  is neglected.
- The cross-section is infinitely rigid in its own plane.
- Suitable for thin beams made of isotropic materials.

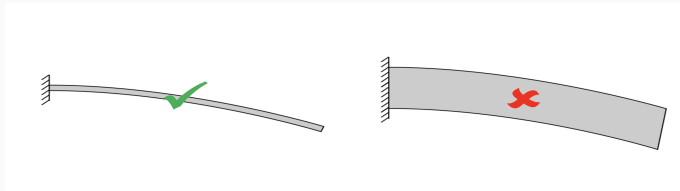
## Thick beam theory



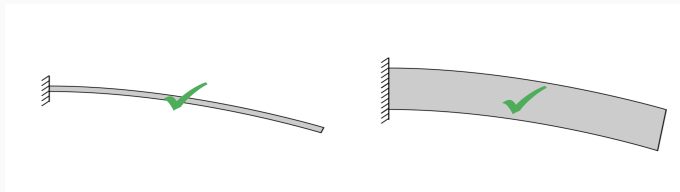
- Cross-sections remain **planar** but **not perpendicular** to the longitudinal axis.
- Shear deformation  $\varepsilon_{12}$  is considered.
- Suitable when shear effects matter, including in composite thick beams.

# Validity range of beam theories

**Euler–Bernoulli:** valid for slender beams:  $h/\ell < 1/100$



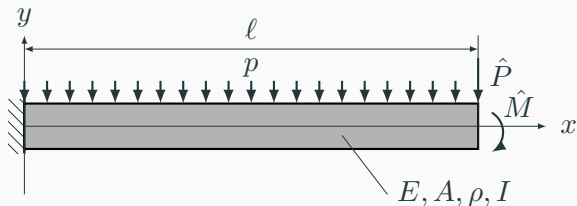
**Timoshenko:** valid for slender beams ( $h/\ell < 1/100$ ) and thick beams ( $h/\ell > 1/10$ )



(Credit: Chatzi and Egger)

# Kinematic assumptions for *planar* Euler-Bernoulli beam

- ① The beam dynamics will be restricted to the  $O(x, y)$  plane.
- ② The beam cross-section remains planar after deformation.
- ③ Lines that are straight and perpendicular to the beam axis remain straight and perpendicular during deformation.



Model parameters:

- $A$  cross-sectional area
- $E$  Young's modulus
- $\rho$  material density
- $I$  moment of inertia
- $\ell$  length

Loads:

- $\hat{M}$  bending moment at free end
- $\hat{P}$  load at free end
- $p$  distributed transversal load

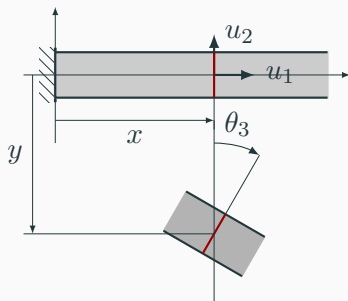
## Variables:

■  $u_1(x, t)$  axial displacement

■  $u_2(x, t)$  transversal displacement

# Displacements field

Introduce an auxiliary variable  $\theta_3(x, t)$  representing the total rotation of the section around the  $Oz$  axis. Rotations are positive in clockwise direction.



- The first Euler-Bernoulli assumption implies

$$u_3 = 0.$$

- The second Euler-Bernoulli assumption implies

$$u_1 = -y\theta_3 \quad (\text{rotation-axial displ.})$$

- The third Euler-Bernoulli assumption implies

$$\theta_3 = \partial_x u_2 \quad (\text{rotation-transversal displ.})$$

**Only one unknown:** transversal displacement  $u_2(x, t)$ .

# Stress, strain and displacement fields

## Strain-displacement relationships

Substituting the displacement field  $\mathbf{u}$  into  $\boldsymbol{\varepsilon} = \nabla \mathbf{u}$  yield to :

$$\varepsilon_{11} = \partial_x u_1 = -y \partial_{xx}^2 u_2$$

$$\varepsilon_{22} = \partial_y u_2 = 0$$

$$\varepsilon_{12} = \partial_x u_2 + \partial_y u_1 = \partial_x u_2 - \theta_3 = 0$$

$$\varepsilon_{33} = \varepsilon_{23} = \varepsilon_{13} = 0$$

## Generalized Hooke's law

Using the linear relationship for homogeneous and isotropic material,  $\boldsymbol{\sigma} = \mathbf{C}\boldsymbol{\varepsilon}$ , is possible to write the stress field in term of the strain field as :

$$\sigma_{11} = (\lambda + 2G)\varepsilon_{11}$$

$$\sigma_{22} = \lambda\varepsilon_{11}$$

$$\sigma_{33} = \lambda\varepsilon_{11}$$

$$\sigma_{12} = \sigma_{13} = \sigma_{23} = 0$$

where  $\lambda = \frac{E}{(1+\nu)(1-2\nu)}$  and the shear modulus  $G = \frac{E}{2(1+\nu)}$  are Lamé constants.

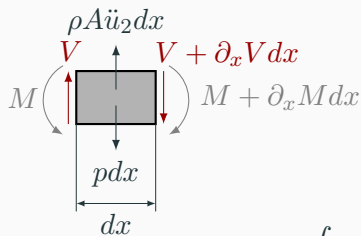
# Dynamic equilibrium equations

The theoretical stresses do not agree well with the experimental measurements.

## Additional assumptions regarding the stress field

$$\sigma_{11} = E\varepsilon_{11}, \quad \text{and} \quad \sigma_{22} = \sigma_{33} = \sigma_{12} = \sigma_{13} = \sigma_{23} = 0.$$

*Justification* : the dimension of the cross-section is much smaller than the beam length.



■ Equilibrium of transverse forces:

$$\partial_x V + p = \rho A \ddot{u}_2$$

■ Equilibrium of moments:

$$\partial_x M + V = 0$$

$$M = - \int_A y \sigma_{11} dA = - \int_A y E \varepsilon_{11} dA = EI \partial_{xx}^2 u_2$$

## Strong form for transversal vibrations for Euler-Bernoulli beam

The strong form consists of finding the transversal displacement  $u_2 \in C^4([0, \ell] \times [0, T])$  such that the following equilibrium equation is satisfied:

$$\partial_{xx}^2 (EI \partial_{xx}^2 u_2) + \rho A \ddot{u}_2 = p.$$

Coupled with four boundary conditions:

$$\begin{aligned} u_2(0, t) &= 0 & \partial_x (-EI \partial_{xx}^2 u_2)(\ell, t) &= \hat{P} \\ \partial_x u_2(0, t) &= 0 & EI \partial_{xx}^2 u_2(\ell, t) &= \hat{M} \end{aligned}$$

and two initial conditions:

$$u_2(x, 0) = u_0(x) \quad \dot{u}_2(x, 0) = v_0(x)$$

## Weak form for transversal vibrations for Euler-Bernoulli beam

Find the transversal displacement  $u_2 \in \mathcal{U}$  such that the following equation is satisfied for every virtual transversal displacement  $\delta u_2 \in \mathcal{V}$ :

$$\int_0^\ell EI \partial_{xx}^2 u_2 \partial_{xx}^2 \delta u_2 dx + \int_0^\ell \rho A \ddot{u}_2 \delta u_2 dx = \int_0^\ell p \delta u_2 dx + \hat{M} \delta u_2'(\ell) + \hat{P} \delta u_2(\ell)$$

$$\mathcal{U} = \{u_2(\cdot, t) \in H^2(]0, \ell[) \mid u_2(0, t) = \partial_x u_2(0, t) = 0 \ \forall t \in ]0, T[ \}$$

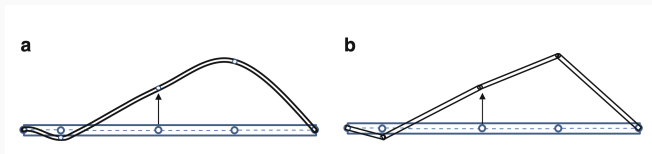
$$\mathcal{V} = \{\delta u_2 \in H^2(]0, \ell[) \mid \delta u_2(0) = \delta u_2'(0) = 0\}$$

The Sobolev space  $H^2(]0, \ell[)$  is defined as:

$$H^2(]0, \ell[) = \left\{ f \in L^2(]0, \ell[) \mid \int_0^\ell (\partial_x f(x))^2 dx < \infty, \int_0^\ell (\partial_{xx}^2 f(x))^2 dx < \infty \right\}.$$

## Approximated transversal displacement

- Euler-Bernoulli theory states that both transversal displacement  $u_2$  and rotation  $\theta_3 = \partial_x u_2$  must be differentiable within finite elements and continuous between elements.
- Shape functions that meet this requirement are said to have **C<sup>1</sup> continuity**.

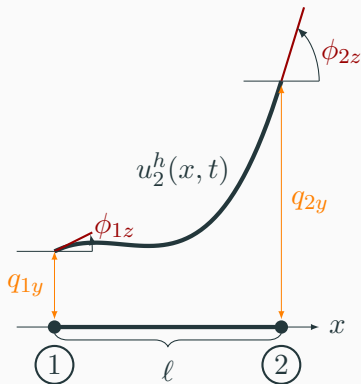


Cubic vs. linear elements. *Credit: [N]*

Assume the approximated transversal displacement to be

$$u_2^h(x, t) = a_3(t)x^3 + a_2(t)x^2 + a_1(t)x + a_0(t)$$

# Nodal degrees of freedom within a beam element



## Parameters shift:

$$a_0, a_1, a_2, a_3 \Rightarrow q_{1y}, q_{2y}, \phi_{1z}, \phi_{2z}$$

A planar beam element has two DOFs per node:

- $q_{1y}$  et  $q_{2y}$ : **nodal displacements** in the transverse direction.
- $\phi_{1z}$  et  $\phi_{2z}$ : **nodal rotations** around the axis normal to the beam plane.

## Hermite $C^1$ shape functions

- Expressing the approximated displacement  $u^h$  as a function of the nodal four DOFs yield to:

$$u_2^h(x, t) = \mathbf{H}(x)\mathbf{q}(t) = [h_1(x), h_2(x), h_3(x), h_4(x)] \begin{bmatrix} q_{1y}(t) \\ \phi_{1z}(t) \\ q_{2y}(t) \\ \phi_{2z}(t) \end{bmatrix}$$

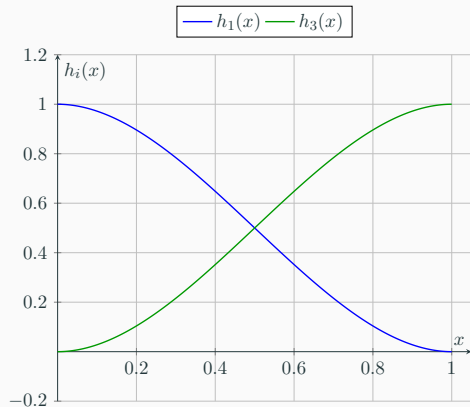
- Cubic local shape functions:

$$\begin{aligned} h_1(x) &= 2(x/\ell)^3 - 3(x/\ell)^2 + 1 & h_3(x) &= 3(x/\ell)^2 - 2(x/\ell)^3 \\ h_2(x) &= x(1 - x/\ell)^2 & h_4(x) &= x(x/\ell)(x/\ell - 1) \end{aligned}$$

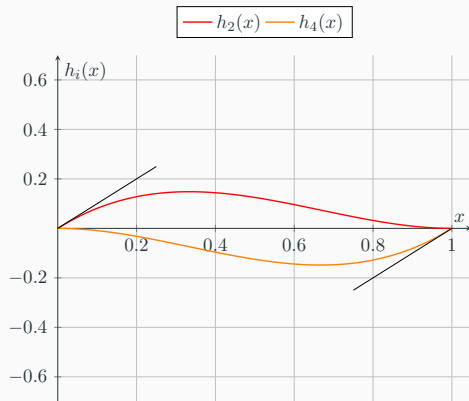
- Virtual displacement approximation follows the same logic:

$$\delta u^h(x) = \mathbf{H}(x)\delta\mathbf{q} \quad \text{where} \quad \delta\mathbf{q} = [\delta u_1, \delta\phi_{1z}, \delta q_{2y}, \delta\phi_{2z}]^T$$

# Hermite $C^1$ shape functions visualizations



Translational shape functions



Rotational shape functions

## Element stiffness matrix

$$\begin{aligned}\mathbf{K} &= \int_0^\ell EI \mathbf{B}^T \mathbf{B} dx \\ &= \int_0^\ell EI \begin{bmatrix} (h_1'')^2 & h_1''h_2'' & h_1''h_3'' & h_1''h_4'' \\ & (h_2'')^2 & h_2''h_3'' & h_2''h_4'' \\ & & (h_3'')^2 & h_3''h_4'' \\ \text{Symm.} & & & (h_4'')^2 \end{bmatrix} dx \\ &= \frac{EI}{\ell^3} \begin{bmatrix} 12 & 6\ell & -12 & 6\ell \\ & 4\ell^2 & -6\ell & 2\ell^2 \\ & & 12 & -6\ell \\ \text{Symm.} & & & 4\ell^2 \end{bmatrix}\end{aligned}$$

The deformation matrix is

$$\mathbf{B} = \frac{d^2 \mathbf{H}}{dx^2} = [h_1''(x) \quad h_2''(x) \quad h_3''(x) \quad h_4''(x)]$$

## Element consistent mass matrix

$$\begin{aligned}\mathbf{M} &= \int_0^\ell \rho A \mathbf{H}^T \mathbf{H} \, dx \\ &= \int_0^\ell \rho A \begin{bmatrix} (h_1)^2 & h_1 h_2 & h_1 h_3 & h_1 h_4 \\ & (h_2)^2 & h_2 h_3 & h_2 h_4 \\ & & (h_3)^2 & h_3 h_4 \\ \text{Symm.} & & & (h_4)^2 \end{bmatrix} dx \\ &= \frac{\rho A \ell}{420} \begin{bmatrix} 156 & 22\ell & 54 & -13\ell \\ & 4\ell^2 & 13\ell & -3\ell^2 \\ & & 156 & -22\ell \\ \text{Symm.} & & & 4\ell^2 \end{bmatrix}\end{aligned}$$

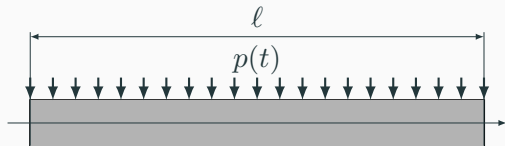
## Element lumped mass matrix

- Assumption that the mass of the structure is lumped at the nodal coordinates where translational displacements are defined.
- Inertial effect associated with any rotational degree of freedom is usually assumed to be zero.
- Recall that  $\rho A$  is the mass per unit length along the beam, then the lumped mass matrix is defined as

$$\mathbf{M} = \frac{\rho A \ell}{2} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

# Applied loads

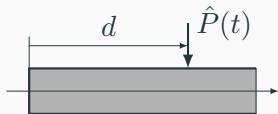
## ■ Uniformly distributed load:



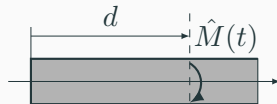
$$\mathbf{r} = \int_0^{\ell} p(t) \mathbf{H}^T dx = \frac{p(t)\ell}{2} \begin{bmatrix} 1 \\ \ell/6 \\ 1 \\ -\ell/6 \end{bmatrix}$$

An uniform transverse load can be replaced by two equivalent transverse nodes loads of value  $(p\ell)/2$  and by two equivalent nodal moments of values  $\pm p\ell^2/12$ .

## ■ Concentrated loads:



$$\mathbf{r}(t) = \hat{P}(t) \mathbf{H}^T(d)$$



$$\mathbf{r}(t) = \hat{M}(t) \frac{d\mathbf{H}^T}{dx}(d)$$

## Post-processing: approximation of bending moment and shear force

- The approximated bending moment along the beam element is

$$M^h(x, t) = EI \partial_{xx}^2 u_2^h(x, t) = EI \mathbf{B}(x) \mathbf{q}(t).$$

- The approximated shear force along the beam element is

$$V^h(x, t) = -\partial_x M^h(x, t) = -EI \frac{d\mathbf{B}}{dx}(x) \mathbf{q}(t).$$

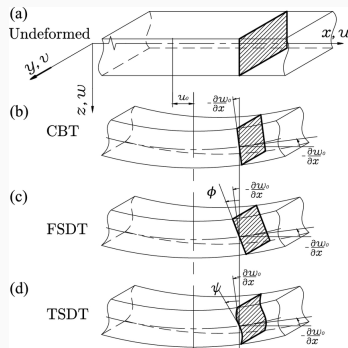
- A Cantilever beam subjected to a downward force
- A clamped-clamped beam (micro transducer) in free vibration.

► [Go to Matlab Drive](#)

# Timoshenko planar beam

---

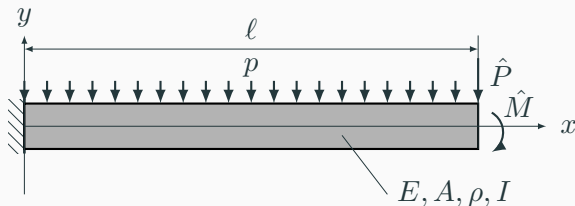
# Higher-order deformation theories for beams



- **Classical beam theory** (Euler-Bernoulli) is a zero-order shear deformation theory.
- **Timoshenko theory** is a first-order shear deformation theory. Cross-sectional planes remain straight but not necessarily perpendicular to the mid-surface.
  - Displacements are linear functions through thickness.
  - In-plane strains are linear in  $z$ .
  - Shear strains are constant in  $z$ .
- **Third-order shear deformation theory**: normal cross-sectional planes to the mid-surface can rotate and deform.
  - Displacements are cubic functions through thickness.
  - In-plane strains are cubic in  $z$ .
  - Shear strains are quadratic in  $z$ .

# Kinematic assumptions for planar Timoshenko beam

- ① The beam dynamics will be restricted to the  $O(x, y)$  plane.
- ② The beam cross-section remains planar after deformation.
- ③ Lines that are straight and perpendicular to the geometrical beam axis remain straight but **not necessarily perpendicular** during deformation.



## Model parameters:

- $A$  cross-sectional area
- $E$  Young's modulus
- $\rho$  material density
- $I$  moment of inertia
- $\ell$  length

## Loads:

- $\hat{M}$  bending moment at free end
- $\hat{P}$  load at free end
- $p$  distributed transversal load

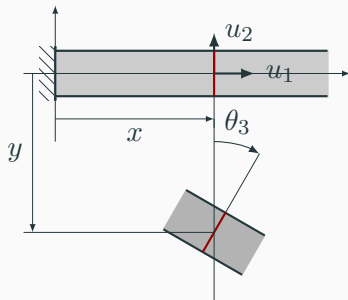
## Variables:

■  $u_1(x, t)$  axial displacement

■  $u_2(x, t)$  transversal displacement

# Displacements field

Introduce an auxiliary variable  $\theta_3(x, t)$  representing the total rotation of the section around the  $Oz$  axis. Rotations are positive in clockwise direction.



- The first Timoshenko assumption implies

$$u_3 = 0.$$

- The second Timoshenko assumption implies

$$u_1 = -y\theta_3 \quad (\text{rotation-axial displ.})$$

**Two unknowns:**  $u_2(x, t)$  transversal displacement and  $\theta_3(x, t)$  rotation of the section around the  $Oz$  axis.

# Stress, strain and displacement fields

## Strain-displacement relationships

Substituting the displacement field  $\mathbf{u}$  into  $\boldsymbol{\varepsilon} = \nabla \mathbf{u}$  yield to :

$$\varepsilon_{11} = \partial_x u_1 = -y \partial_{xx}^2 u_2$$

$$\varepsilon_{22} = \partial_y u_2$$

$$\varepsilon_{12} = \partial_x u_2 + \partial_y u_1 = \partial_x u_2 - \theta_3$$

$$\varepsilon_{33} = \varepsilon_{23} = \varepsilon_{13} = 0$$

## Generalized Hooke's law

Using the linear relationship for homogeneous and isotropic material,  $\boldsymbol{\sigma} = \mathbf{C}\boldsymbol{\varepsilon}$ , is possible to write the stress field in term of the strain field as :

$$\sigma_{11} = (\lambda + 2G)\varepsilon_{11}$$

$$\sigma_{12} = 2G\varepsilon_{12}$$

$$\sigma_{22} = \sigma_{33} = \lambda\varepsilon_{11}$$

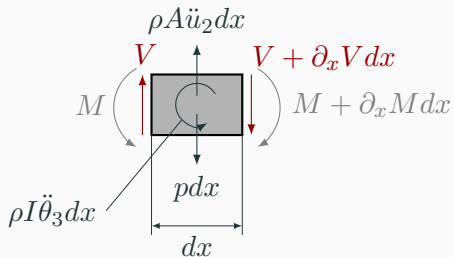
$$\sigma_{13} = \sigma_{23} = 0$$

where  $\lambda = \frac{E}{(1+\nu)(1-2\nu)}$  and the shear modulus  $G = \frac{E}{2(1+\nu)}$  are Lamé constants.

# Dynamic equilibrium equations

## Additional assumptions regarding the stress field

$$\sigma_{11} = E\varepsilon_{11}, \quad \sigma_{12} = 2G\varepsilon_{12} \quad \text{and} \quad \sigma_{22} = \sigma_{33} = \sigma_{13} = \sigma_{23} = 0.$$



- Equilibrium of transverse forces:

$$\partial_x V + p = \rho A \ddot{u}_2$$

- Equilibrium of moments:

$$\partial_x M + V = \rho I \ddot{\theta}_3$$

$$V = \int_A \sigma_{12} dA = kGA\varepsilon_{12} = kGA(\partial_x u_2 - \theta_3)$$

$$M = - \int_A y \sigma_{11} dA = - \int_A y E \varepsilon_{11} dA = EI \partial_x \theta_3$$

## Strong form for transversal vibrations for Timoshenko beam

The weak formulation of the problem consists of finding the transversal displacement  $u_2 \in C^2([0, \ell] \times [0, T])$  and the rotation  $\theta_3 \in C^2([0, \ell] \times [0, T])$  such that the following equilibrium equations, boundary and initial conditions are satisfied.

$$\begin{aligned}\partial_x(kGA(\partial_x u_2 - \theta_3)) + p &= \rho A \ddot{u}_2 \\ \partial_x(EI \partial_x \theta_3) + kGA(\partial_x u_2 - \theta_3) &= \rho I \ddot{\theta}_3\end{aligned}$$

In matrix form:  $\nabla_\sigma^T \mathbf{C} \nabla_u \mathbf{u} + \mathbf{f} = \mathbf{M} \ddot{\mathbf{u}}$ , where

$$\underbrace{\begin{bmatrix} \partial_x & 0 \\ 1 & \partial_x \end{bmatrix}}_{\nabla_\sigma^T} \underbrace{\begin{bmatrix} kGA & 0 \\ 0 & EI \end{bmatrix}}_{\mathbf{C}} \underbrace{\begin{bmatrix} \partial_x & -1 \\ 0 & \partial_x \end{bmatrix}}_{\nabla_u} \underbrace{\begin{bmatrix} u_2 \\ \theta_3 \end{bmatrix}}_{\mathbf{u}} + \underbrace{\begin{bmatrix} p \\ 0 \end{bmatrix}}_{\mathbf{f}} = \underbrace{\begin{bmatrix} \rho A & 0 \\ 0 & \rho I \end{bmatrix}}_{\mathbf{M}} \underbrace{\begin{bmatrix} \ddot{u}_2 \\ \ddot{\theta}_3 \end{bmatrix}}_{\ddot{\mathbf{u}}}$$

## Boundary conditions

$$u_2(0, t) = 0$$

$$\theta_3(0, t) = 0$$

$$kGA(\partial_x u_2(\ell, t) - \theta_3(\ell, t)) = \hat{P}$$

$$EI\partial_x \theta_3(\ell, t) = \hat{M}$$

In matrix form: let  $\hat{\mathbf{f}} = [\hat{P}, \hat{M}]^T$  then

$$\mathbf{u}(0, t) = \mathbf{0}$$

$$\mathbf{C}\nabla_u \mathbf{u}(\ell, t) = \hat{\mathbf{f}}$$

## Initial conditions

$$u_2(x, 0) = u_0(x)$$

$$\theta_3(x, 0) = \theta_0(x)$$

$$\dot{u}_2(x, 0) = v_0(x)$$

$$\dot{\theta}_3(x, 0) = \phi_0(x)$$

In matrix form:  $\mathbf{u}_0 = [u_0, \theta_0]^T$  and  $\mathbf{v}_0 = [v_0, \phi_0]^T$  then

$$\mathbf{u}(x, 0) = \mathbf{u}_0$$

$$\dot{\mathbf{u}}(x, 0) = \mathbf{v}_0$$

## Weak form for transversal vibrations for Timoshenko beam

The weak formulation of the problem consists in finding the unknown function  $\mathbf{u} \in \mathcal{U}$  which satisfies the equation  $\forall \delta \mathbf{u} \in \mathcal{V}$ :

$$\int_0^\ell (\nabla_u \delta \mathbf{u})^T \mathbf{C} (\nabla_u \mathbf{u}) dx + \int_0^\ell \delta \mathbf{u}^T \mathbf{M} \ddot{\mathbf{u}} dx = \int_0^\ell \delta \mathbf{u}^T \mathbf{f} dx + \delta \mathbf{u}^T(\ell) \hat{\mathbf{f}}.$$

The function spaces  $\mathcal{U}$  and  $\mathcal{V}$  are defined as follows

$$\mathcal{U} = \left\{ \mathbf{u} = \{u_2, \theta_3\}^T \mid u_2(\cdot, t) \in H^1(]0, \ell[); \theta_3(\cdot, t) \in H^1(]0, \ell[); u_2(0, t) = \theta_3(0, t) = 0 \right\},$$
$$\mathcal{V} = \left\{ \delta \mathbf{u} = \{\delta u_2, \delta \theta_3\}^T \mid \delta u_2 \in H^1(]0, \ell[); \delta \theta_3 \in H^1(]0, \ell[); \delta u_2(0) = \delta \theta_3(0) = 0 \right\}.$$

# Displacement approximation

- We approximate the generalized displacement using the ansatz:

$$\mathbf{u}^h(\mathbf{x}, t) = \mathbf{H}(\mathbf{x})\mathbf{q}(t) \quad \text{and} \quad \delta\mathbf{u}^h(\mathbf{x}) = \mathbf{H}(\mathbf{x})\delta\mathbf{q}$$

- **Cubic shape functions** are used to approximate the transversal displacement and **linear shape functions** for the rotation field:

$$\mathbf{H} = \begin{bmatrix} h_1^c & h_2^c & h_3^c & h_4^c \\ 0 & h_1^l & 0 & h_2^l \end{bmatrix}$$

$$h_1^c(x) = 2(x/\ell)^3 - 3(x/\ell)^2 + 1$$

$$h_3^c(x) = 3(x/\ell)^2 - 2(x/\ell)$$

$$h_2^c(x) = x(1 - x/\ell)^2$$

$$h_4^c(x) = x(x/\ell)(x/\ell - 1)$$

$$h_1^l(x) = 1 - x/\ell$$

$$h_2^l(x) = x/\ell$$

- $\mathbf{q}(t)$  is a vector of (*unknown*) nodal displacements and  $\delta\mathbf{q}$  is a vector of constants:

$$\mathbf{q}(t) = \begin{bmatrix} q_{1y}(t) & \phi_{1z}(t) & q_{2y}(t) & \phi_{2z}(t) \end{bmatrix}^T \quad \text{and} \quad \delta\mathbf{q} = \begin{bmatrix} \delta q_{1y} & \delta\phi_{1z} & \delta q_{2y} & \delta\phi_{2z} \end{bmatrix}^T$$

## Deformation, stiffness, mass matrices and loads vector

- $\mathbf{B} = \nabla_u \mathbf{H}$  is the deformation matrix:

$$\mathbf{B} = \begin{bmatrix} \partial_x h_1^c & \partial_x h_2^c - h_1^l & \partial_x h_3^c & \partial_x h_4^c - h_2^l \\ 0 & \partial_x h_1^l & 0 & \partial_x h_2^l \end{bmatrix}$$

- The stiffness  $\mathbf{K}$  and mass  $\mathbf{Ma}$  matrices and load vector  $\mathbf{r}$  are defined as

$$\mathbf{K} = \int_0^\ell \mathbf{B}^T \mathbf{C} \mathbf{B} \, dx,$$

$$\mathbf{Ma} = \int_0^\ell \mathbf{H}^T \mathbf{M} \mathbf{H} \, dx,$$

$$\mathbf{r}(t) = \int_0^\ell \mathbf{H}^T \mathbf{f} \, dx + \mathbf{H}^T(\ell) \hat{\mathbf{f}}.$$

- The semi-discrete equations of motion are given by

$$\mathbf{Ma} \ddot{\mathbf{q}}(t) + \mathbf{K} \mathbf{q}(t) = \mathbf{r}(t).$$

# MATLAB example - modal analysis of a Timoshenko beam

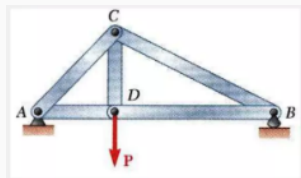
► [Go to Matlab Drive](#)

# Plane frames

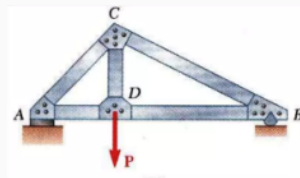
---

# What is a plane frame?

- Structure composed of oriented thin beam elements, connected by welding, and carrying **transversal and axial forces** that all lies in a common plane.
- Both forces and moments can be transmitted between members.
- Loads are acting only in the common plane of the structure and they must be applied at the joints.



Plane truss

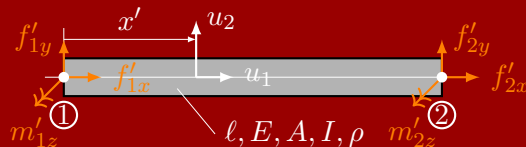


Plane frame

Truss + Thin beams = Frame

## Axial loads in thin beam element

The inclusion of axial forces in a flexural beam element requires a *superposition* of bar and beam elements:



Differential equations governing the dynamics:

$$\partial_{x'x'}^2 (EI \partial_{x'x'}^2 u_2(x', t)) + \rho A \ddot{u}_2(x', t) = 0$$

$$EA \partial_{x'x'}^2 u_1(x', t) = \rho A \ddot{u}_1(x', t)$$

*Axial effects are subsequently incorporated into the beam element formulation, unless specified otherwise.*

## Axial displacement in thin beam element

A beam element now has three degrees of freedom per node:  $q'_{ix}$ ,  $q'_{iy}$ , and  $\phi'_{iz}$ .

- Displacements approximation:

$$u^h(x', t) = \mathbf{H}(x') \mathbf{q}_{loc}(t) = [h_1(x') \ h_2(x') \ h_3(x') \ h_4(x') \ h_5(x') \ h_6(x')] \begin{bmatrix} q'_{1x}(t) \\ q'_{1y}(t) \\ \phi'_{1z}(t) \\ q'_{2x}(t) \\ q'_{2y}(t) \\ \phi'_{2z}(t) \end{bmatrix}$$

- Local shape functions:

$$h_1(x') = 1 - x'/\ell$$

$$h_2(x') = 2(x'/\ell)^3 - 3(x'/\ell)^2 + 1$$

$$h_3(x') = x'(1 - x'/\ell)^2$$

$$h_4(x') = x'/\ell$$

$$h_5(x') = 3(x'/\ell)^2 - 2(x'/\ell)^3$$

$$h_6(x') = x'(x'/\ell)(x'/\ell - 1)$$

# Discretization of thin beam including axial effects

- Element stiffness matrix in local coordinates:

$$\mathbf{K}_{loc} = \frac{EI}{\ell^3} \begin{bmatrix} A\ell^2/I & 0 & 0 & -A\ell^2/I & 0 & 0 \\ & 12 & 6\ell & 0 & -12 & 6\ell \\ & & 4\ell^2 & 0 & -6\ell & 2\ell^2 \\ & & & A\ell^2/I & 0 & 0 \\ & & & & 12 & -6\ell \\ sym. & & & & & 4\ell^2 \end{bmatrix}$$

- Element consistent mass matrix in local coordinates:

$$\mathbf{M}_{loc} = \frac{\rho A \ell}{420} \begin{bmatrix} 140 & 0 & 0 & 70 & 0 & 0 \\ & 0 & 156 & 22\ell & 0 & 54 \\ & & & 4\ell^2 & 0 & 13\ell \\ & & & & 70 & 0 \\ & & & & & 0 \\ & & & & & 156 \\ sym. & & & & & & -22\ell \\ & & & & & & & 4\ell^2 \end{bmatrix}$$

# Arbitrarily oriented thin beam element including axial effects

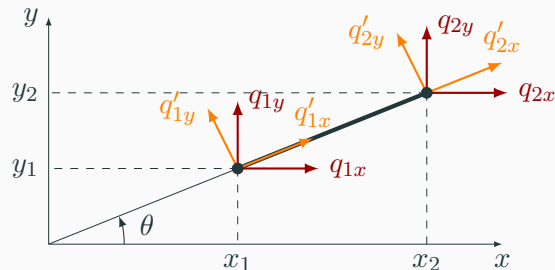
- Displacements in local coordinates:

$$\mathbf{q}_{loc} = [q'_{1x} \quad q'_{1y} \quad \phi'_{1z} \quad q'_{2x} \quad q'_{2y} \quad \phi'_{2z}]^T.$$

- Displacements in global coordinates:

$$\mathbf{q} = [q_{1x} \quad q_{1y} \quad \phi_{1z} \quad q_{2x} \quad q_{2y} \quad \phi_{2z}]^T.$$

- Relation between local and global displacements:



$$\underbrace{\begin{bmatrix} q'_{1x} \\ q'_{1y} \\ \phi'_{1z} \\ q'_{2x} \\ q'_{2y} \\ \phi'_{2z} \end{bmatrix}}_{\mathbf{q}_{loc}} = \underbrace{\begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 & 0 & 0 & 0 \\ -\sin(\theta) & \cos(\theta) & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos(\theta) & \sin(\theta) & 0 \\ 0 & 0 & 0 & -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}}_{\mathbf{T}} \underbrace{\begin{bmatrix} q_{1x} \\ q_{1y} \\ \phi_{1z} \\ q_{2x} \\ q_{2y} \\ \phi_{2z} \end{bmatrix}}_{\mathbf{q}}$$

# Discretization of arbitrarily oriented beam including axial effects

- Element stiffness matrix in global coordinates:

$$\mathbf{K} = \mathbf{T}^T \mathbf{K}_{loc} \mathbf{T} = \frac{E}{\ell} \begin{bmatrix} \frac{A\ell^3 C^2 + 12IS^2}{\ell^2} & \frac{(-12I + A\ell^2)CS}{\ell^2} & \frac{-6IS}{\ell} & -\frac{A\ell^3 C^2 + 12IS^2}{\ell^2} & \frac{(12I - A\ell^2)CS}{\ell^2} & \frac{-6IS}{\ell} \\ & \frac{12IC^2 + A\ell^2 S^2}{\ell^2} & \frac{6IC}{\ell} & \frac{(12I - A\ell^2)CS}{\ell^2} & -\frac{12IC^2 + A\ell^2 S^2}{\ell^2} & \frac{6IC}{\ell} \\ & & 4I & \frac{6IS}{\ell} & \frac{-6IC}{\ell} & 2I \\ & & & \frac{A\ell^3 C^2 + 12IS^2}{\ell^2} & \frac{(-12I + A\ell^2)CS}{\ell^2} & \frac{6IS}{\ell} \\ & & & & \frac{12IC^2 + A\ell^2 S^2}{\ell^2} & \frac{-6IC}{\ell} \\ & & & & & 4I \end{bmatrix}$$

where  $C = \cos(\theta)$  and  $S = \sin(\theta)$ .

- Element consistent mass matrix in global coordinates:

$$\mathbf{M} = \mathbf{T}^T \mathbf{M}_{loc} \mathbf{T}$$

# Applied loads

- Element applied loads vector in local coordinates:

$$\mathbf{f}_{loc}(t) = \begin{bmatrix} f'_{1x}(t) \\ f'_{1y}(t) \\ m'_{1z}(t) \\ f'_{2x}(t) \\ f'_{2y}(t) \\ m'_{2z}(t) \end{bmatrix}$$

- Element applied loads vector in global coordinates:

$$\mathbf{f} = \mathbf{T}^T \mathbf{f}_{loc} = \begin{bmatrix} \cos(\theta)f'_{1x} + \sin(\theta)f'_{1y} \\ -\sin(\theta)f'_{1x} + \cos(\theta)f'_{1y} \\ m'_{1z} \\ \cos(\theta)f'_{2x} + \sin(\theta)f'_{2y} \\ -\sin(\theta)f'_{2x} + \cos(\theta)f'_{2y} \\ m'_{2z} \end{bmatrix}$$

# Assembly of stiffness and mass matrices and loads vector

Given a 2d frame structure made of  $m$  oriented beams,  $n$  nodes, and 3 DOFs per node:

## 1. Element quantities:

- For each beam  $e$ , compute the element quantities global coordinates:

$${}^e\mathbf{K} = {}^e\mathbf{T}^T {}^e\mathbf{K}_{loc} {}^e\mathbf{T}$$

$${}^e\mathbf{M} = {}^e\mathbf{T}^T {}^e\mathbf{M}_{loc} {}^e\mathbf{T}$$

$${}^e\mathbf{f} = {}^e\mathbf{T}^T {}^e\mathbf{f}_{loc}$$

## 2. Global assembly:

- Initialize global stiffness matrix  $\mathbf{K}$  and global mass matrix  $\mathbf{M}$  of size  $3n \times 3n$ ,
- Initialize global loads vector  $\mathbf{f}$  of size  $3n \times 1$ ,
- Assemble each  ${}^e\mathbf{K}$ ,  ${}^e\mathbf{M}$  and  ${}^e\mathbf{f}$  for  $e = 1, \dots, m$ , into  $\mathbf{K}$ ,  $\mathbf{M}$  and  $\mathbf{f}$  respectively using element connectivity.

# MATLAB example - frame 2d in free vibration

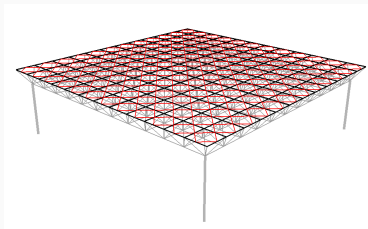
► [Go to Matlab Drive](#)

# Plane grids

---

# What is a grid?

- A grid is a structure composed of oriented planar beams subjected to **perpendicular loading**, producing significant bending effects.
- Beams are connected by welding, so forces and moments can be transmitted between members.
- The axial effect of displacement is ignored for now.
- Grids commonly model floors, roofs, bridge decks, and similar structures.



Shaft + thin beams = Grid



# Degrees of freedom identification

## ■ Planar frame under action of loads in the plane of the structure:

Components required to describe the displacements of joint  $i$  are:

- $q_{ix}$  translation in the  $x$  direction,
- $q_{iy}$  translation in the  $y$  direction,
- $\phi_{iz}$  bending, rotation about the  $z$  axis.

## ■ Planar grids loaded perpendicularly to the plane of the structure:

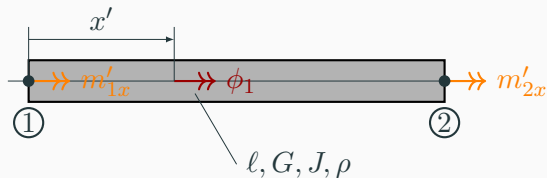
Components required to describe the displacements of joint  $i$  are:

- $q_{iz}$  translation in the  $z$  direction,
- $\phi_{ix}$  torsion, rotation about the  $x$  axis .
- $\phi_{iy}$  bending, rotation about the  $z$  axis.

## Shaft element

The development of a shaft finite element is very similar to the development of a bar finite element, where

- the axial displacement  $u_1$  is replaced by the angular rotation  $\phi_1$ ,
- the axial nodal forces  $f'_{ix}$  are replaced by nodal torque  $m'_{ix}$ ,
- the element tensile stiffness  $AE/\ell$  is replaced by the torsional stiffness  $GJ/\ell$ .



- $G$  shear modulus of the material
- $J$  polar moment of inertia of the cross-section.
- $\rho$  material density
- $\ell$  length
- $\phi_1$  angular rotation
- $x'$  (local) axial coordinate

## Equation of motion for non-oriented shaft element

Differential equation governing the dynamics:

$$GJ\partial_{x'x'}^2\phi_1(x',t) = \rho J\ddot{\phi}_1(x',t)$$

- Angular rotation (twisting) approximation:

$$\phi_1^h(x',t) = \mathbf{H}(x')\mathbf{q}_{loc}(t) = \begin{bmatrix} h_1(x') & h_2(x') \end{bmatrix} \begin{bmatrix} \phi'_{1x}(t) \\ \phi'_{2x}(t) \end{bmatrix}$$

- Linear local shape functions:

$$h_1(x') = 1 - \frac{x'}{\ell} \quad \text{and} \quad h_2(x') = \frac{x'}{\ell}$$

- Semi-discrete weak form:

$$\delta\mathbf{q}_{loc}^T [\mathbf{M}_{loc}\ddot{\mathbf{q}}_{loc}(t) + \mathbf{K}_{loc}\mathbf{q}_{loc}(t) - \mathbf{f}_{loc}(t)] = 0$$

- Element stiffness matrix in local coordinates:

$$\mathbf{K}_{loc} = \int_0^\ell GJ \frac{d\mathbf{H}^T}{dx'} \frac{d\mathbf{H}}{dx'} dx' = \int_0^\ell GJ \begin{bmatrix} (h'_1)^2 & h'_1 h'_2 \\ h'_2 h'_1 & (h'_2)^2 \end{bmatrix} dx' = \frac{GJ}{\ell} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

- Element consistent mass matrix in local coordinates:

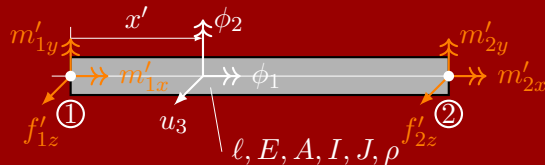
$$\mathbf{M}_{loc} = \int_0^\ell \rho J \mathbf{H}^T \mathbf{H} dx' = \int_0^\ell \rho J \begin{bmatrix} (h_1)^2 & h_1 h_2 \\ h_2 h_1 & (h_2)^2 \end{bmatrix} dx' = \frac{\rho J \ell}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

- Element applied loads vector in local coordinates:

$$\mathbf{f}_{loc}(t) = \begin{bmatrix} h_1(0) \\ h_2(0) \end{bmatrix} m'_{1x}(t) + \begin{bmatrix} h_1(\ell) \\ h_2(\ell) \end{bmatrix} m'_{2x}(t) = \begin{bmatrix} m'_{1x}(t) \\ m'_{2x}(t) \end{bmatrix}$$

## Torsional effects in thin beam element

The inclusion of torsional stiffness in a thin beam element, to model a typical element in a planar grid frame, requires a *superposition* of shaft and beam elements:



Differential equations governing the dynamics:

$$\partial_{x'x'}^2 (EI \partial_{x'x'}^2 u_3(x', t)) + \rho A \ddot{u}_3(x', t) = 0$$

$$GJ \partial_{x'x'}^2 \phi_1(x', t) = \rho J \ddot{\phi}_1(x', t)$$

## Torsional effects in thin beam element

Each grid element now has three degrees of freedom per node:  $\phi'_{ix}$ ,  $\phi'_{iy}$  and  $q'_{iz}$ .

- Displacements approximation:

$$u^h(x', t) = \mathbf{H}(x') \mathbf{q}_{loc}(t) = [h_1(x') \ h_2(x') \ h_3(x') \ h_4(x') \ h_5(x') \ h_6(x')] \begin{bmatrix} \phi'_{1x}(t) \\ \phi'_{1y}(t) \\ q'_{1z}(t) \\ \phi'_{2x}(t) \\ \phi'_{2y}(t) \\ q'_{2z}(t) \end{bmatrix}$$

- Local shape functions:

$$h_1(x') = 1 - x'/\ell$$

$$h_2(x') = x'(1 - x'/\ell)^2$$

$$h_3(x') = 2(x'/\ell)^3 - 3(x'/\ell)^2 + 1$$

$$h_4(x') = x'/\ell$$

$$h_5(x') = x'(x'/\ell)(x'/\ell - 1)$$

$$h_6(x') = 3(x'/\ell)^2 - 2(x'/\ell)^3$$

# Discretization of thin beam with torsional effects

- Element stiffness matrix in local coordinates:

$$\mathbf{K}_{loc} = \frac{EI}{\ell^3} \begin{bmatrix} GJ\ell^2/EI & 0 & 0 & -GJ\ell^2/EI & 0 & 0 \\ & 4\ell^2 & -6\ell & 0 & 2\ell^2 & 6\ell \\ & & 12 & 0 & -6\ell & -12 \\ & & & GJ\ell^2/EI & 0 & 0 \\ & & & & 4\ell^2 & 6\ell \\ sym. & & & & & 12 \end{bmatrix}$$

- Element consistent mass matrix in local coordinates:

$$\mathbf{M}_{loc} = \frac{\rho A \ell}{420} \begin{bmatrix} 140J/A & 0 & 0 & 70J/A & 0 & 0 \\ & 0 & 4\ell^2 & 22\ell & 0 & -3\ell^2 & 13\ell \\ & & & 156 & 0 & -13\ell & 54 \\ & & & & 140J/A & 0 & 0 \\ & & & & & 4\ell^2 & -22\ell \\ sym. & & & & & & 156 \end{bmatrix}$$

# Arbitrarily oriented thin beam element with torsional effects

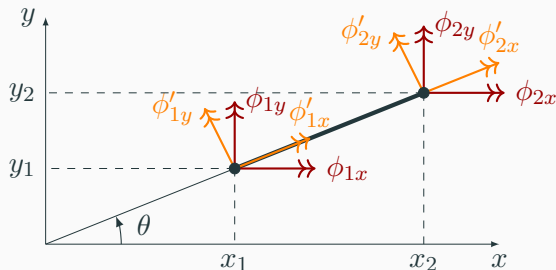
- Displacements in local coordinates:

$$\mathbf{q}_{loc} = [\phi'_{1x} \quad \phi'_{1y} \quad q'_{1z} \quad \phi'_{2x} \quad \phi'_{2y} \quad q'_{2z}]^T.$$

- Displacements in global coordinates:

$$\mathbf{q} = [\phi_{1x} \quad \phi_{1y} \quad q_{1z} \quad \phi_{2x} \quad \phi_{2y} \quad q_{2z}]^T.$$

- Relation between local and global displacements:



$$\underbrace{\begin{bmatrix} \phi'_{1x} \\ \phi'_{1y} \\ q'_{1z} \\ \phi'_{2x} \\ \phi'_{2y} \\ q'_{2z} \end{bmatrix}}_{\mathbf{q}_{loc}} = \underbrace{\begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 & 0 & 0 & 0 \\ -\sin(\theta) & \cos(\theta) & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos(\theta) & \sin(\theta) & 0 \\ 0 & 0 & 0 & -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}}_{\mathbf{T}} \underbrace{\begin{bmatrix} \phi_{1x} \\ \phi_{1y} \\ q_{1z} \\ \phi_{2x} \\ \phi_{2y} \\ q_{2z} \end{bmatrix}}_{\mathbf{q}}$$

# Discretization of arbitrarily oriented thin beam

- Element stiffness matrix in global coordinates:

$$\mathbf{K} = \mathbf{T}^T \mathbf{K}_{loc} \mathbf{T}$$

where  $C = \cos(\theta)$  and  $S = \sin(\theta)$ .

- Element consistent mass matrix in global coordinates:

$$\mathbf{M} = \mathbf{T}^T \mathbf{M}_{loc} \mathbf{T}$$

- Element applied loads vector in global coordinates:

$$\mathbf{f} = \mathbf{T}^T \mathbf{f}_{loc}$$

# MATLAB example - dynamic response of a 2d grid

► [Go to Matlab Drive](#)

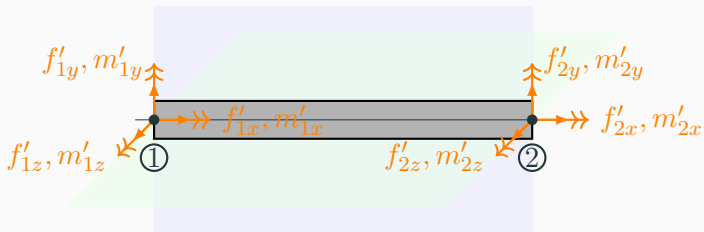
# Three-dimensional frames

---

## Example of a three-dimensional structure composed of thin beams



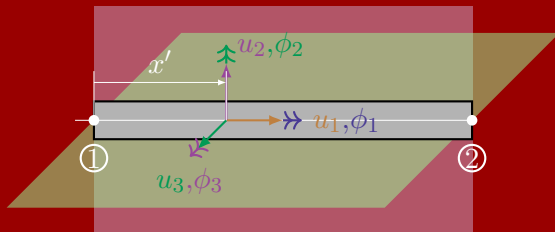
Credit: [N]



Three-dimensional thin beams are uniaxial (slender) element that can support:

- axial loads  $f'_{ix}$ ,
- torsional loads  $m'_{ix}$ ,
- bending in the  $x' - y'$  plane:  $f'_{iy}$  and  $m'_{iz}$ ,
- bending in the  $x' - z'$  plane:  $f'_{iz}$  and  $m'_{iy}$ .

# Differential equations governing the dynamics of thin 3d beams



$$EA\partial_{x'x'}^2 u_1(x', t) = \rho A \ddot{u}_1(x', t)$$

$$\partial_{x'x'}^2 (EI_z \partial_{x'x'}^2 u_2(x', t)) + \rho A \ddot{u}_2(x', t) = 0$$

$$GJ\partial_{x'x'}^2 \phi_1(x', t) = \rho J \ddot{\phi}_1(x', t)$$

$$\partial_{x'x'}^2 (EI_y \partial_{x'x'}^2 u_3(x', t)) + \rho A \ddot{u}_3(x', t) = 0$$

- $E$  Young's modulus of the material.
- $A$  cross-sectional area.
- $\rho$  material density.
- $I_y$  and  $I_z$  are the second moments of inertia with respect to  $Oy$  and  $Oz$ .
- $J$  polar moment of inertia of the cross-section.

# Displacements discretization

Total of six nodal generalized displacements at each unconstrained joint:

- three translation components  $q'_{ix}$ ,  $q'_{iy}$  and  $q'_{iz}$  along the  $x$ ,  $y$  and  $z$  axes,
- three rotational components  $\phi'_{ix}$ ,  $\phi'_{iy}$  and  $\phi'_{iz}$  about the  $x$ ,  $y$  and  $z$  axes.

$$u^h(x', t) = \mathbf{H}(x') \mathbf{q}_{loc}(t)$$

$$\mathbf{q}_{loc}(t) = \begin{bmatrix} q'_{1x}(t) \\ q'_{1y}(t) \\ q'_{1z}(t) \\ \phi'_{1x}(t) \\ \phi'_{1y}(t) \\ \phi'_{1z}(t) \\ q'_{2x}(t) \\ q'_{2y}(t) \\ q'_{2z}(t) \\ \phi'_{2x}(t) \\ \phi'_{2y}(t) \\ \phi'_{2z}(t) \end{bmatrix}$$

$$h_1(x') = 1 - x'/\ell$$

$$h_2(x') = 2(x'/\ell)^3 - 3(x'/\ell)^2 + 1$$

$$h_3(x') = 2(x'/\ell)^3 - 3(x'/\ell)^2 + 1$$

$$h_4(x') = 1 - x'/\ell$$

$$h_5(x') = x'(1 - x'/\ell)^2$$

$$h_6(x') = x'(1 - x'/\ell)^2$$

$$h_7(x') = x'/\ell$$

$$h_8(x') = 3(x'/\ell)^2 - 2(x'/\ell)^3$$

$$h_9(x') = 3(x'/\ell)^2 - 2(x'/\ell)^3$$

$$h_{10}(x') = x'/\ell$$

$$h_{11}(x') = x'(x'/\ell)(x'/\ell - 1)$$

$$h_{12}(x') = x'(x'/\ell)(x'/\ell - 1)$$

## Element stiffness matrix in local coordinates

The stiffness matrix for a three-dimensional uniform beam element is written as the superposition of the **axial stiffness** matrix, the **torsional stiffness** matrix and the **flexural stiffness** matrix:

$$\mathbf{K}_{loc} = \begin{bmatrix} \frac{EA}{\ell} & 0 & 0 & 0 & 0 & 0 & -\frac{EA}{\ell} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{12EI_z}{\ell^3} & 0 & 0 & 0 & \frac{6EI_z}{\ell^2} & 0 & -\frac{12EI_z}{\ell^3} & 0 & 0 & 0 & \frac{6EI_z}{\ell^2} \\ 0 & 0 & \frac{12EI_y}{\ell^3} & 0 & -\frac{6EI_y}{\ell^2} & 0 & 0 & 0 & -\frac{12EI_y}{\ell^3} & 0 & -\frac{6EI_y}{\ell^2} & 0 \\ 0 & 0 & \frac{12EI_y}{\ell^3} & 0 & -\frac{6EI_y}{\ell^2} & 0 & 0 & 0 & -\frac{12EI_y}{\ell^3} & 0 & -\frac{6EI_y}{\ell^2} & 0 \\ 0 & 0 & 0 & \frac{GJ}{\ell} & 0 & 0 & 0 & 0 & 0 & -\frac{GJ}{\ell} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{4EI_y}{\ell} & 0 & 0 & 0 & \frac{6EI_y}{\ell^2} & 0 & \frac{2EI_y}{\ell} & 0 \\ 0 & 0 & 0 & 0 & \frac{4EI_y}{\ell} & 0 & 0 & 0 & \frac{6EI_y}{\ell^2} & 0 & \frac{2EI_y}{\ell} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{4EI_z}{\ell} & 0 & -\frac{6EI_z}{\ell^2} & 0 & 0 & 0 & \frac{2EI_z}{\ell} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{EA}{\ell} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{12EI_z}{\ell^3} & 0 & 0 & 0 & -\frac{6EI_z}{\ell^2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{12EI_z}{\ell^3} & 0 & 0 & 0 & -\frac{6EI_z}{\ell^2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{12EI_y}{\ell^3} & 0 & \frac{6EI_y}{\ell^2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{12EI_y}{\ell^3} & 0 & \frac{6EI_y}{\ell^2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{GJ}{\ell} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4EI_y}{\ell} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4EI_z}{\ell} \end{bmatrix}$$

*sym.*

## Element consistent mass matrix in local coordinates

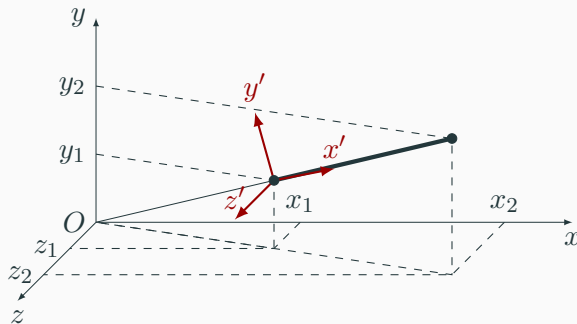
The consistent mass matrix for a three-dimensional uniform beam element is written as the superposition of the **axial mass** matrix, the **torsional mass** matrix and the **flexural mass** matrix:

$$\mathbf{M}_{loc} = \frac{\rho A \ell}{420} \begin{bmatrix} 140 & 0 & 0 & 0 & 0 & 0 & 70 & 0 & 0 & 0 & 0 & 0 \\ & 156 & 0 & 0 & 0 & 22\ell & 0 & 54 & 0 & 0 & 0 & -13\ell \\ & & 156 & 0 & -22\ell & 0 & 0 & 0 & 54 & 0 & 13\ell & 0 \\ & & & \frac{140J}{A} & 0 & 0 & 0 & 0 & 0 & \frac{70J}{A} & 0 & 0 \\ & & & & 4\ell^2 & 0 & 0 & 0 & -13\ell & 0 & -3\ell^2 & 0 \\ & & & & & 4\ell^2 & 0 & 13\ell & 0 & 0 & 0 & -3\ell^2 \\ & & & & & & 140 & 0 & 0 & 0 & 0 & 0 \\ & & & & & & & 156 & 0 & 0 & 0 & -22\ell \\ & & & & & & & & 156 & 0 & 22\ell & 0 \\ & & & & & & & & & \frac{140J}{A} & 0 & 0 \\ & & & & & & & & & & 4\ell^2 & 0 \\ & & & & & & & & & & & 4\ell^2 \end{bmatrix}$$

*sym.*

# Transformation of coordinates

- The stiffness and mass matrices are defined in the local coordinate system  $x'$ ,  $y'$  and  $z'$  fixed to the beam segment.
- To assemble the global stiffness and mass matrices, these local matrices must be transformed into the global coordinate system  $x$ ,  $y$ ,  $z$ .



## Direction cosines

- The relationship between local and global coordinates is:

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \underbrace{\begin{bmatrix} \cos(x'x) & \cos(x'y) & \cos(x'z) \\ \cos(y'x) & \cos(y'y) & \cos(y'z) \\ \cos(z'x) & \cos(z'y) & \cos(z'z) \end{bmatrix}}_{\mathbf{T}'} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

- The local  $x'$  is given by:

$$Ox' = [l_x, l_y, l_z]$$

where

$$l_x = \cos(x'x) = \frac{x_2 - x_1}{e_\ell}, \quad l_y = \cos(x'y) = \frac{y_2 - y_1}{e_\ell}, \quad l_z = \cos(x'z) = \frac{z_2 - z_1}{e_\ell},$$

$$e_\ell = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

## Direction cosines

- The local  $y'$  axis is chosen so that it lies in the local  $x' - y'$  plane and it is perpendicular to  $x'$ :

$$Oy' = \frac{1}{d} [l_y, l_x, 0]$$

thus

$$\cos(y'x) = \frac{l_y}{d}, \quad \cos(y'y) = \frac{l_x}{d}, \quad \cos(y'z) = 0, \quad d = \sqrt{l_x^2 + l_y^2}.$$

- The local  $z'$  axis is chosen so that  $Oz' = Ox' \times Oy'$ :

$$Oz' = \frac{1}{d} [-l_x l_z, -l_y l_z, 0]$$

thus

$$\cos(z'x) = -\frac{l_x l_z}{d}, \quad \cos(z'y) = -\frac{l_y l_z}{d}, \quad \cos(z'z) = d.$$

# Global displacements for oriented three-dimensional beam

$$\underbrace{\begin{bmatrix} q'_{1x}(t) \\ q'_{1y}(t) \\ q'_{1z}(t) \\ \phi'_{1x}(t) \\ \phi'_{1y}(t) \\ \phi'_{1z}(t) \\ q'_{2x}(t) \\ q'_{2y}(t) \\ q'_{2z}(t) \\ \phi'_{2x}(t) \\ \phi'_{2y}(t) \\ \phi'_{2z}(t) \end{bmatrix}}_{\mathbf{q}_{loc}(t)} = \underbrace{\begin{bmatrix} \mathbf{T}' & & \\ & \mathbf{T}' & \\ & & \mathbf{T}' \end{bmatrix}}_{\mathbf{T}} \underbrace{\begin{bmatrix} q_{1x}(t) \\ q_{1y}(t) \\ q_{1z}(t) \\ \phi_{1x}(t) \\ \phi_{1y}(t) \\ \phi_{1z}(t) \\ q_{2x}(t) \\ q_{2y}(t) \\ q_{2z}(t) \\ \phi_{2x}(t) \\ \phi_{2y}(t) \\ \phi_{2z}(t) \end{bmatrix}}_{\mathbf{q}(t)}$$

